

Homework 7 Lyapunov's method

The given non-linear differential equation

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -\frac{3x_1 + x_3}{x_2 + 1} \\ -\frac{x_2 + 2x_3}{x_2 + 3} \\ x_1 \end{bmatrix}$$

part (a) use Lyapunov's indirect method

```
clearvars, clc;
syms x1 x2 x3
RHS = [-(3*x1+x3)/(x2+1);
       -(x2+2*x3)/(x2+3);
       x1]
```

RHS =

$$\begin{pmatrix} -\frac{3x_1 + x_3}{x_2 + 1} \\ -\frac{x_2 + 2x_3}{x_2 + 3} \\ x_1 \end{pmatrix}$$

```
xdot = RHS;
x = [x1; x2; x3];
J = jacobian(RHS, x);

A = subs(J, {x1, x2, x3}, {0, 0, 0});

eigenvalues = eig(A);
ev = double(eigenvalues)
```

```
ev = 3x1
     -0.3333
     -2.6180
     -0.3820
```

```
real_parts = real(ev)
```

```
real_parts = 3x1
     -0.3333
     -2.6180
     -0.3820
```

```
if all(real_parts < 0)
    disp('All real parts < 0, system is asymptotically stable at
equilibrium')
```

```

else
    disp('exist real parts >= 0')
end

```

All real parts < 0, system is asymptotically stable at equilibrium

part (b) initial response

here

```

f = @(t, x) [-(3*x(1) + x(3)) / (x(2) + 1);
             -(x(2) + 2*x(3)) / (x(2) + 3);
             x(1)]

```

```

f = function_handle with value:
    @(t,x)[-(3*x(1)+x(3))/(x(2)+1);-(x(2)+2*x(3))/(x(2)+3);x(1)]

```

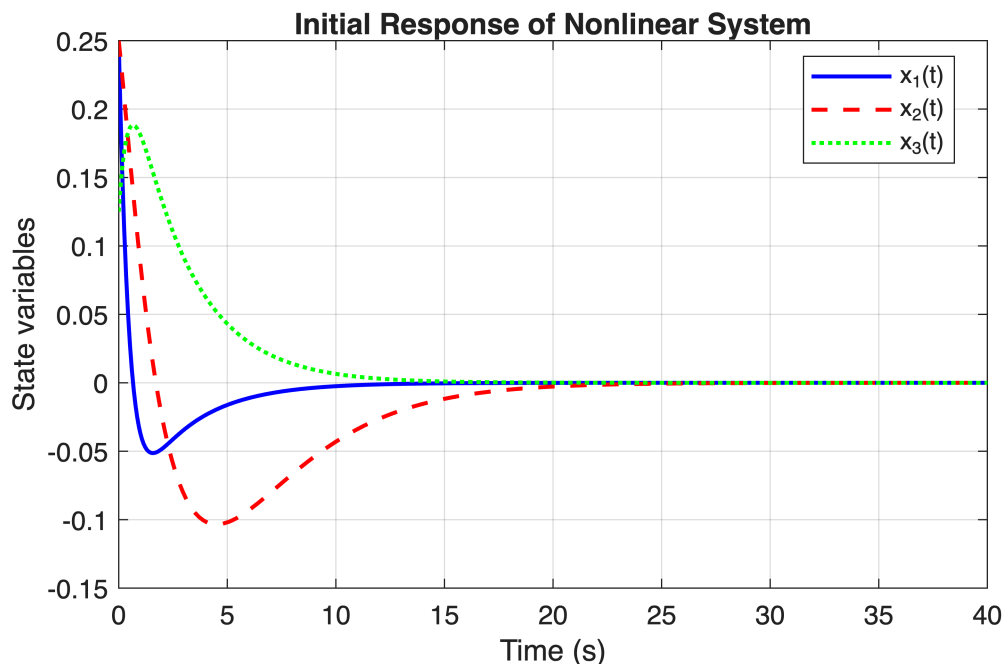
```

x0 = [0.25; 0.25; 0.125];
tspan = [0, 40];
[t, x] = ode45(f, tspan, x0);

figure;
plot(t, x(:,1), 'b-', 'LineWidth', 1.5); hold on;
plot(t, x(:,2), 'r--', 'LineWidth', 1.5);
plot(t, x(:,3), 'g:', 'LineWidth', 1.5);

xlabel('Time (s)');
ylabel('State variables');
title('Initial Response of Nonlinear System');
legend('x_1(t)', 'x_2(t)', 'x_3(t)', 'Location', 'best');
grid on;

```



part (c) find the lyapunov's function and plot time history

The lyapunov's equation is:

$$A^T P + P A = -Q$$

where $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The P can be found from the lyapunov's equation, The candidate function V is defined as

$$V(x) = x^T P x$$

$$\text{Then } \dot{V}(x) = \dot{x}^T P x + x^T P \dot{x} = f(x)^T P x + x^T P f(x)$$

where $\dot{x} = f(x)$

```
Q = eye(3);  
%solve A'P + PA = -Q  
P = lyap(A', -Q); % MATLAB 内置：解 A'P + PA = -Q, 注意符号  
disp(eig(P));
```

```
-6.1132  
-0.9123  
-0.1849
```

```
f = @(t, x) [-(3*x(1) + x(3)) / (x(2) + 1);  
             -(x(2) + 2*x(3)) / (x(2) + 3);  
             x(1)];
```

```
x0 = [0.25; 0.25; 0.125];  
tspan = [0, 40];  
[t, x] = ode45(f, tspan, x0);
```

```
% Vdot = f(x)'*P*x + x'*P*f(x) = 2 * x'*P*f(x)
```

```
Vdot = zeros(length(t), 1);
```

```
V = zeros(length(t), 1);
```

```
for i = 1:length(t)  
    xi = x(i,:); % 当前状态列向量  
    fi = f(0, xi); % 非线性 f(x)  
    V(i) = xi' * P * xi; % Lyapunov 函数值  
    Vdot(i) = fi' * P * xi + xi' * P * fi; % dV/dt  
end
```

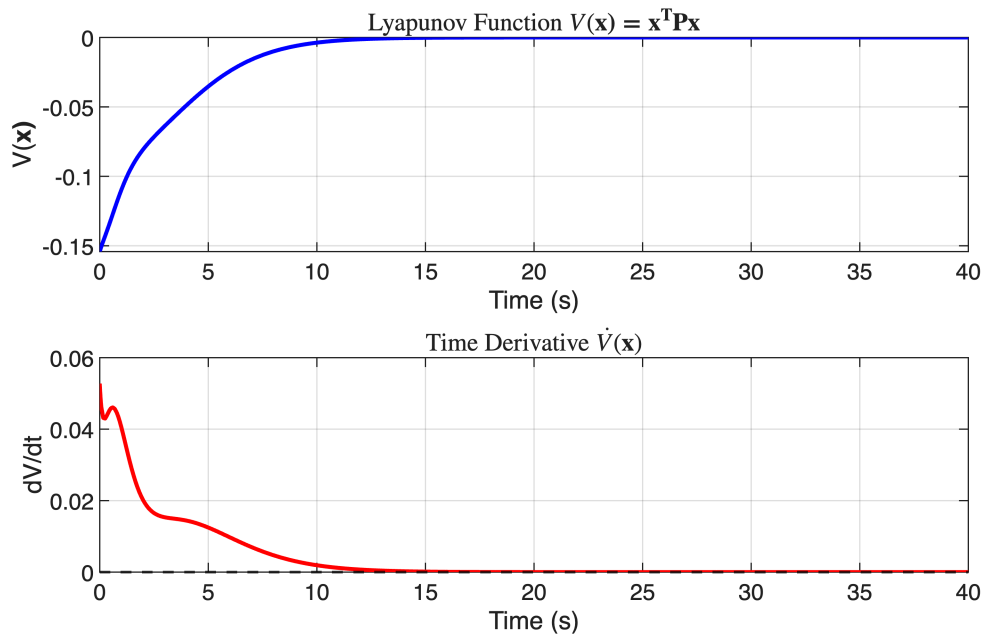
```
figure;  
subplot(2,1,1);  
plot(t, V, 'b-', 'LineWidth', 1.5);  
xlabel('Time (s)'); ylabel('V(\bf{x})');
```

```

title('Lyapunov Function  $V(\mathbf{x}) = \mathbf{x}^T P \mathbf{x}$ '
', 'Interpreter', 'latex');
grid on;

subplot(2,1,2);
plot(t, Vdot, 'r-', 'LineWidth', 1.5);
yline(0, 'k--', 'LineWidth', 1);
xlabel('Time (s)'); ylabel('dV/dt');
title('Time Derivative  $\dot{V}(\mathbf{x})$ ', 'Interpreter', 'latex');
grid on;

```



part (d) explore different initial condition

i n i t i a l c o n d i t	distance from the origin	behavior

o n		
[0.25, 0.25, 0.125] ^T	(x) negative all the time	converge and stable
[0.5, 0.5, 0.25] ^T	(x) negative almost all the time	converge and stable
[1.0, 1.0, 0.5] ^T	(x) negative most time	
[2.0, 0.5, 1.0] ^T	(x) > 1.0 at the beginning	probabiliy stable
[0.1, -0.1, 0.05] ^T	(x) negative all the time	converge and stable

```
clearvars, clc;

%% 系统与 Lyapunov 函数
f = @(t, x) [-(3*x(1) + x(3)) / (x(2) + 1);
             -(x(2) + 2*x(3)) / (x(2) + 3);
             x(1)];

A = [-3,    0,   -1;
      0,   -1/3, -2/3;
      1,    0,    0];

P    = lyap(A', eye(3));
tspan = [0, 40];

%% 多组初始条件
X0 = [0.25,  0.25,  0.125; % b 问原始条件 (近)
      0.5,   0.5,   0.25;  % 稍远
      1.0,   1.0,   0.5;   % 更远
      2.0,   0.5,   1.0;   % 较大初值
      0.1,  -0.1,   0.05]; % 含负值

colors = ['b', 'r', 'g', 'm', 'c'];
labels = {'x_0=[0.25, 0.25, 0.125]', ...
          'x_0=[0.5,  0.5,  0.25 ]', ...
          'x_0=[1.0,  1.0,  0.5  ]', ...
          'x_0=[2.0,  0.5,  1.0  ]', ...
          'x_0=[0.1, -0.1,  0.05 ]'};

options = odeset('RelTol', 1e-6, 'AbsTol', 1e-9, ...
                 'Events', @singularity_event);

figure('Position', [100 100 1400 700]);

for k = 1:size(X0, 1)
```

```

x0 = X0(k,:);

%% 求解 (加 try-catch 防止崩溃)
try
    [t, x] = ode45(f, tspan, x0, options);
catch
    warning('初始条件 %s 积分失败, 跳过', labels{k});
    continue;
end

%% 计算 V 和 Vdot
V = zeros(length(t), 1);
Vdot = zeros(length(t), 1);
for i = 1:length(t)
    xi = x(i,:);
    fi = f(0, xi);
    V(i) = xi' * P * xi;
    Vdot(i) = fi' * P * xi + xi' * P * fi;
end

%% 绘图
% 状态变量
subplot(3, size(X0,1), k)
plot(t, x(:,1), 'b-', t, x(:,2), 'r--', t, x(:,3), 'g:', 'LineWidth', 1.2)
title(labels{k}, 'FontSize', 8)
xlabel('$t$ (s)', 'Interpreter', 'latex')
ylabel('$x_i(t)$', 'Interpreter', 'latex')
legend('$x_1$', '$x_2$', '$x_3$', 'Interpreter', 'latex', ...
    'Location', 'best', 'FontSize', 7)
grid on;

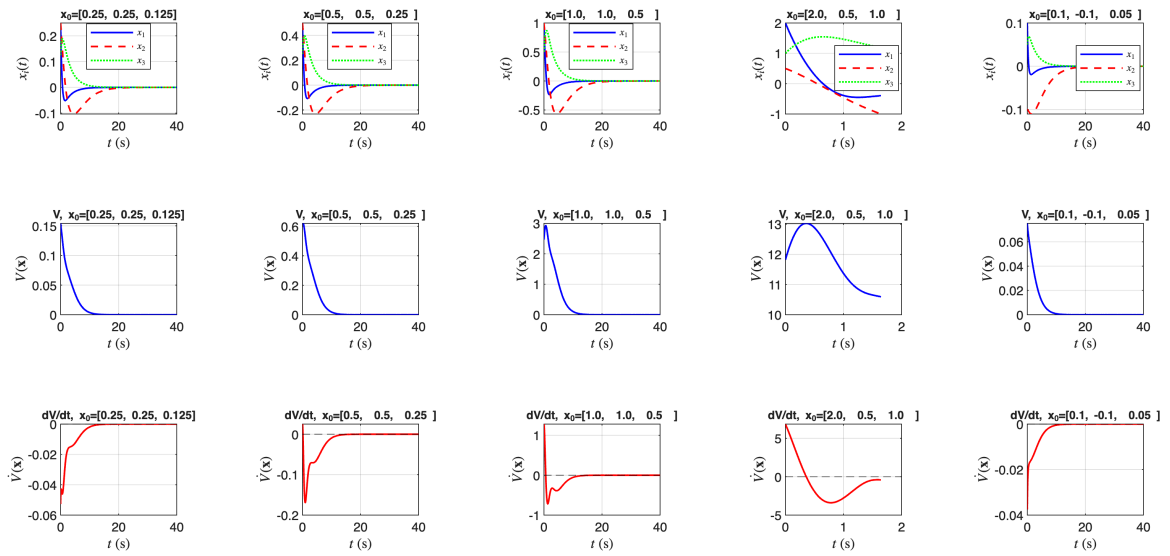
% V
subplot(3, size(X0,1), k + size(X0,1))
plot(t, V, 'b-', 'LineWidth', 1.2)
title(['V, ' labels{k}], 'FontSize', 8)
xlabel('$t$ (s)', 'Interpreter', 'latex')
ylabel('$V(\mathbf{x})$', 'Interpreter', 'latex')
grid on;

% Vdot
subplot(3, size(X0,1), k + 2*size(X0,1))
plot(t, Vdot, 'r-', 'LineWidth', 1.2)
yline(0, 'k--')
title(['dV/dt, ' labels{k}], 'FontSize', 8)
xlabel('$t$ (s)', 'Interpreter', 'latex')
ylabel('$\dot{V}(\mathbf{x})$', 'Interpreter', 'latex')
grid on;
end

sgtitle('Part (d): Different Initial Conditions')

```

Part (d): Different Initial Conditions



%% 奇点检测函数

```
function [value, isterminal, direction] = singularity_event(~, x)
    value      = min(abs(x(2) + 1), abs(x(2) + 3)) - 1e-3;
    isterminal = 1;
    direction  = -1;
end
```